

Exploratory Data Analysis (EDA) Summary

1. Introduction

This report presents an exploratory data analysis of the customer delinquency dataset. The main objective is to understand the quality and structure of the data, identify important patterns and unusual observations, and find factors that may be linked to delinquent accounts. The analysis also focuses on missing values and inconsistencies so they can be addressed before building a predictive model.

2. Dataset Overview

The dataset contains 500 customer records and 19 variables covering customer details, financial information, credit behavior, payment history, and delinquency status. The main variables include income, loan balance, debt-to-income ratio, credit score, credit utilization, missed payments, employment status, account tenure, credit card type, and monthly payment history. The dataset contains both numerical and categorical variables, along with a binary target variable indicating whether an account is delinquent.

Key dataset attributes:

- **Number of records:** 500
- **Key variables:** Customer ID, Age, Location, Income, Loan Balance, Debt-to-Income Ratio, Credit Score, Credit Utilization, Missed Payments, Account Tenure, Employment Status, Credit Card Type, Month 1–Month 6 Payment Status, and Delinquent Account.
- **Data types:** Numerical variables include Age, Income, Loan Balance, Debt-to-Income Ratio, Credit Score, Credit Utilization, Missed Payments, and Account Tenure. Categorical variables include Location, Employment Status, Credit Card Type, and monthly payment status. `Delinquent_Account` is the binary target variable.

3. Missing Data Analysis

Missing values were reviewed to understand their potential impact on the delinquency analysis. Most of the dataset is complete, with missing values found mainly in **Income, Loan Balance, and Credit Score**. Rather than removing customers with missing information, appropriate methods were selected to retain the records and reduce the impact of missing data.

Missing Data Issue	Handling Method	Justification
Income (39 missing values, 7.8%)	Synthetic values using a normal distribution	This preserves the dataset size while generating realistic values based on the observed income distribution.
Loan Balance (29 missing values, 5.8%)	Median Imputation	The median is less affected by extreme values and provides a robust replacement for missing loan balances.
Credit Score (2 missing values, 0.4%)	Median Imputation	Only a very small number of values are missing, so simple median imputation is sufficient.

Key missing data findings:

- **Variables with missing values:** Income - 39 values (7.8%), Loan Balance - 29 values (5.8%), and Credit Score - 2 values (0.4%).
- **Missing data treatment:** Missing Income values were replaced with realistic synthetic values generated using a normal distribution assumption and constrained to the observed income range. Missing Loan Balance and Credit Score values were handled using median imputation, as the median provides a simple and robust replacement for numerical financial data.

4. Key Findings and Risk Indicators

The analysis shows that delinquency is influenced by a combination of financial and payment-related factors rather than a single variable. Credit Utilization and Debt-to-Income Ratio are slightly higher among delinquent customers, suggesting that greater credit usage and debt burden may be linked to higher delinquency risk. Payment history also appears useful, as customers with recent Late payments show higher delinquency rates in some months.

Key findings:

- **Correlations observed between Key variables:** Delinquent customers show slightly higher Credit Utilization (about 0.51 vs. 0.49) and Debt-to-Income Ratio (about 0.31 vs. 0.30) compared with non-delinquent customers. However, the differences are small and should be further validated before these variables are treated as strong predictors of delinquency.
- **Unexpected Anomalies:** Delinquent customers have a slightly higher average Credit Score and Income than non-delinquent customers, which is unexpected and should be investigated further. In addition, four Credit Utilization values are above 100%, and Missed_Payments does not always match the number of “Missed” entries across the monthly payment-history fields. These issues should be validated before final model development.

5. AI & GenAI Usage

Generative AI was used throughout the analysis to help review the dataset, identify important patterns, suggest suitable ways to handle missing values, and highlight potential risk factors. The AI-generated suggestions were then reviewed against the dataset findings before being included in the report.

Example AI prompts used:

- “Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.”
- “Suggest an imputation strategy for missing values in this dataset based on industry best practices.”
- “Generate realistic synthetic income values for missing entries using normal distribution assumptions.”
- “Detect patterns and risk factors that influence delinquency and highlight any unexpected findings that may require further investigation.”

6. Conclusion & Next Steps

Overall, the dataset provides useful information for understanding customer delinquency, but a few data-quality issues need to be addressed before building the predictive model. The analysis suggests that credit utilization, debt-to-income ratio, payment history, employment status, and account tenure may be useful indicators of delinquency risk. At the same

time, some unexpected patterns, such as higher average credit scores among delinquent customers, need further investigation.

Recommended next steps are to validate the unusual values and inconsistencies, standardize categorical data, complete the missing-value treatment, and create useful features from the monthly payment history. The next stage should then focus on handling the class imbalance, selecting the most useful predictors, and evaluating different predictive models using appropriate measures such as precision, recall, F1-score, and ROC-AUC.